

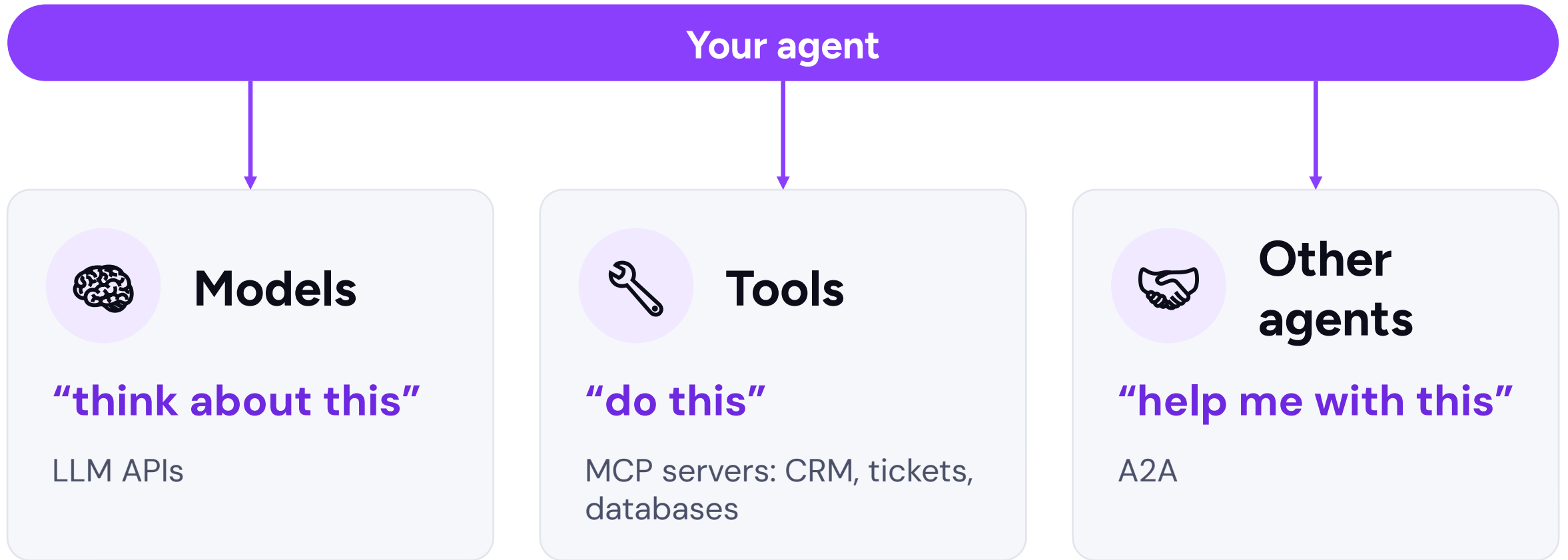


Production-Ready Agent Infrastructure & Agent Gateways

An introduction to agentgateway

Chee Hong Ngu · AAILF meetup · October 2026

Agents talk to everything



Every arrow is a place where things go wrong.

What goes wrong in production



Keys everywhere

Every agent holds its own provider and tool keys



Uncapped spend

A loop runs all night, you find out from the invoice



Unaudited tool access

An agent that should read can also delete



No visibility

Something happened, nobody can say who or why



Outages

A provider is slow or down and the agent hangs

What is agentgateway?



agentgateway in one slide



“One high-performance gateway for service, LLM, and MCP traffic.”

Apache 2.0

Open source, built in Rust

Aug 2025

Created by Solo.io, joined the Linux Foundation

June 2026

Became the Agentic AI Foundation's 4th hosted project

300+

active contributors from 60+ organizations

Contributors include Microsoft, AWS, Adobe, Cisco, Salesforce, Solo.io

Why not a normal API gateway?



A normal API gateway sees

a request

a URL

a status code

An agent gateway also sees

which model

how many tokens

what's in the prompt

which
tool

who's asking

And it speaks the agent protocols natively: MCP and A2A.

Architecture: one front door

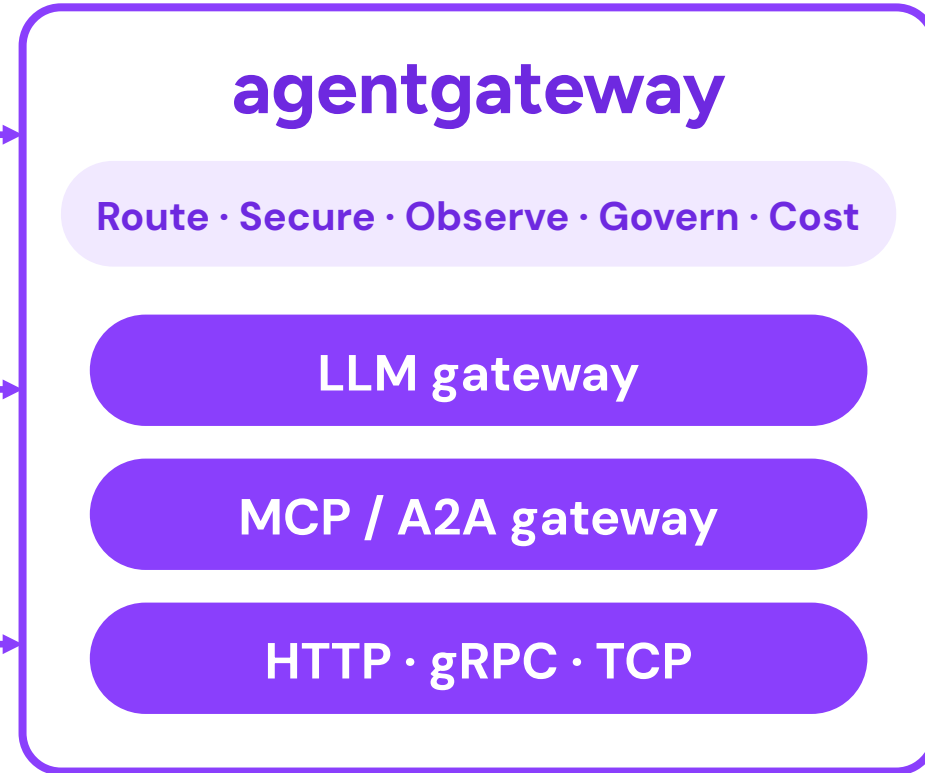


SOURCES

AI agents

Apps & clients

CLI / IDE



DESTINATIONS

LLM providers

MCP tool servers

Other agents

Your APIs

Configured by: a YAML file · the built-in UI · Kubernetes (Gateway API)

What can it do?

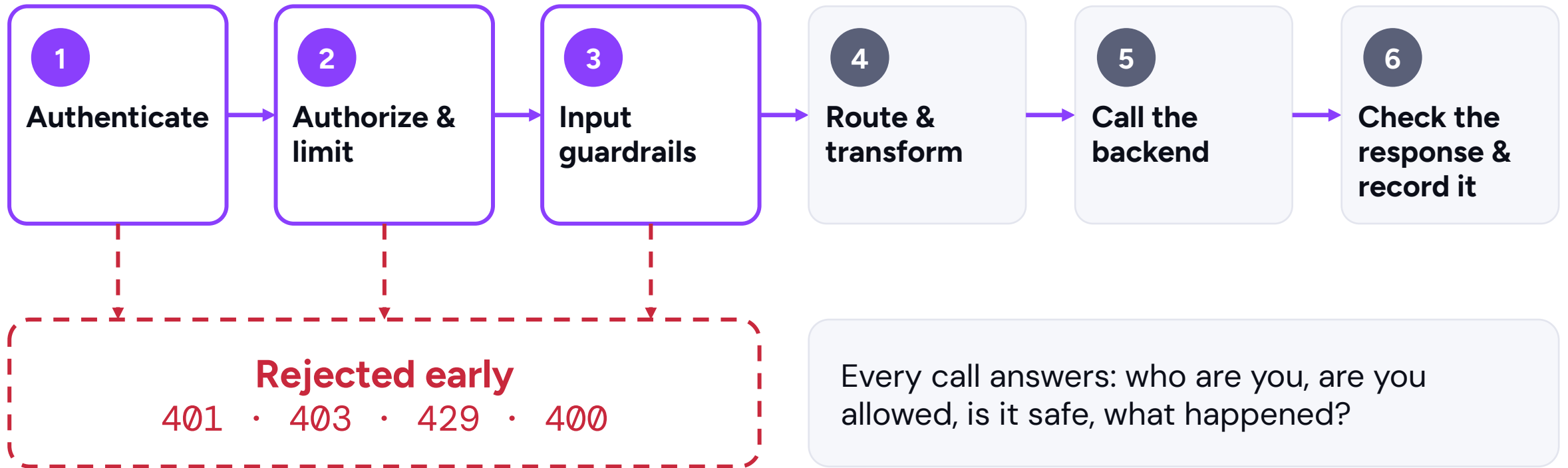
LLM

MCP & A2A

Security

Observability

How a request flows



A mental model, not the exact internal order.

LLM gateway



One API in front of many LLM providers

OpenAI

Anthropic

Gemini

Bedrock

Azure

Vertex AI

Ollama

Mistral

and more

- **Virtual models:** one model name, routed behind the scenes (e.g. by caller)
- **Virtual API keys:** per team or agent, with allowed models
- **Budgets:** cap tokens or dollars per key; audit or block
- **Token-based rate limits**

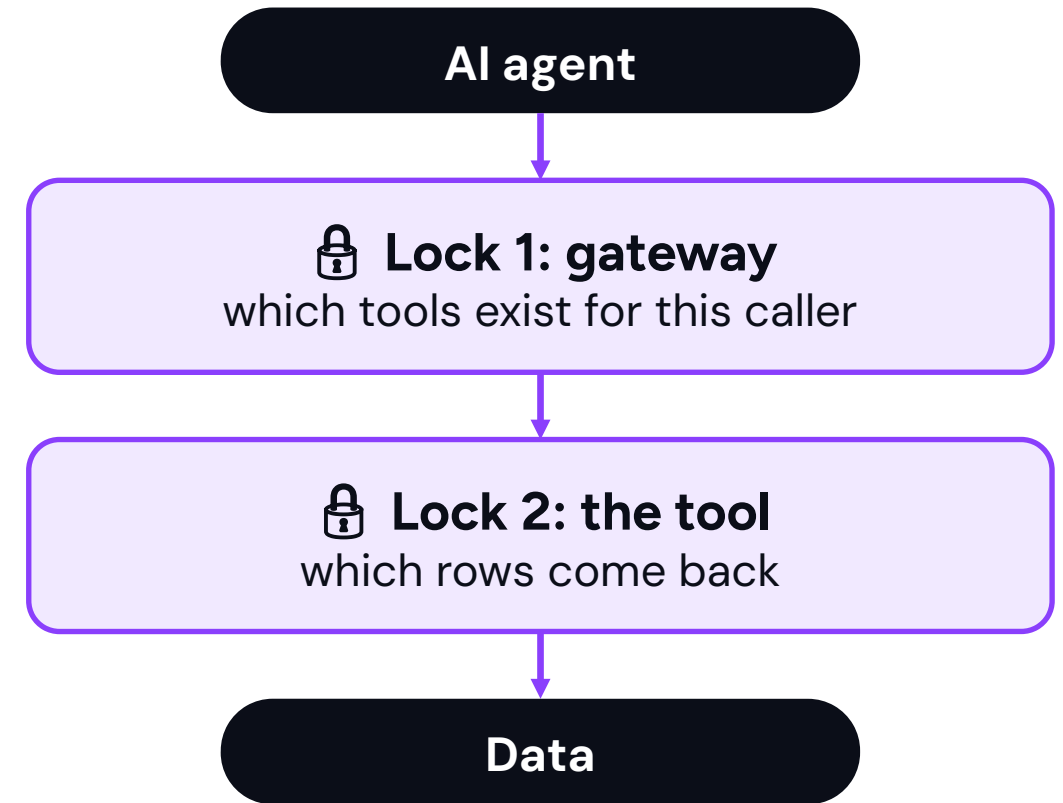
- **Cost** per key and model (estimated from public prices)
- **Load balancing and failover** across models
- Chat, Responses, Messages, embeddings **APIs**

▶ In the demo: Virtual API keys · virtual model routing · LLM budget (USD per key)



MCP & A2A gateway

- Many tool servers behind one `/mcp` endpoint (local or remote)
- Turn REST APIs (OpenAPI) into MCP tools
- Checks who is calling (JWT / OAuth)
- Per-tool rules: callers only see the tools they're allowed
- Proxies agent-to-agent (A2A) traffic, incl. AWS Bedrock AgentCore
- A2A agent cards list each agent's skills; the gateway rewrites the card URL to itself, with per-key auth and rate limits



The gateway forwards a verified identity and strips the user's token.

▶ In the demo: OpenAPI to MCP · MCP authentication (JWT) · MCP authorization · A2A API key + rate limit

Security & guardrails



Who & what is allowed

- JWT, API keys, basic auth, OIDC browser login
- Rules for HTTP, network, MCP, or your own service (external authz)
- CORS & CSRF
- TLS and mutual TLS to backends
- Signs in to AWS, Google Cloud, Azure for you

Guardrails

- **Block** sensitive data on the way in (regex: emails, card numbers, secrets)
- **Mask** personal data on the way out
- Plug in OpenAI moderation, AWS Bedrock Guardrails, Azure AI Content Safety, Google Model Armor, or your own webhook

▶ In the demo: Prompt guardrail: reject · Response guardrail: mask · MCP authorization on arguments

Observability & operations



See everything

Request logs incl. prompts (database, OTLP), Prometheus metrics, OpenTelemetry traces

Plug into your tools

Langfuse · LangSmith · Braintrust · Arize Phoenix · Datadog

Built-in web UI

Logs, costs, analytics, playgrounds, and the full config

Change without downtime

Edit the config, it reloads live. Run as one binary, Docker, or on Kubernetes

▶ In the demo: Request logs + cost tracking · hot reload (live config edit)

New in agentgateway 1.6



Per-key rate limits

each key gets its own bucket, so one noisy caller is limited alone

[▶ In the demo](#)

MCP size limit

oversized MCP requests are refused with a 413 (not shown today)

MCP server name & instructions

the gateway can introduce itself to agents with a name and instructions

Response idle timeout

a stuck model can't freeze the caller (set as a safety net today)

Built-in model catalog

cost tracking for common public models out of the box

Hashed API keys in the UI

stored hashed in the standalone web UI

OIDC sign-in for standalone

log in to the standalone gateway with your identity provider

Config, not code (1/2): models & guardrails



```
llm:
  models:
    - name: upstream/glm-5.3-flash
      provider: {reference: upstream} # any OpenAI-compatible API
  virtualModels:
    - name: assistant
      routing:
        conditional:
          targets:
            - model: upstream/glm-5.3-flash
  policies:
    apiKey:
      keys:
        - key: agw_pp_erlich
          metadata: {name: pp-erlich, team: pied-piper}
          allowedModels: ["assistant"]
          budgets:
            - limit: {unit: USD, amount: 0.0002}
              window: {rolling: 1m}
              onBudgetExceeded: Block
    guardrails:
      request:
        - regex:
            action: reject
            rules:
              - pattern: 'sk-live-[A-Za-z0-9]{8,}'
      response:
        - regex:
            action: mask
            rules:
              - builtin: phoneNumber
```

WHAT EACH PART POWERS

`virtualModels`

Routing, any beat (Logs show requested vs sent)

`budgets`

Beat 6: Erlich gets a 429

`guardrails.request regex`

Beat 2: Dinesh's key, 400

`guardrails.response mask`

Beat 4: Jared's phone number masked

Trimmed from the demo's config.yaml (the real budget also has a name; more built-in guardrail patterns).



Config, not code (2/2): tools & agents

```
mcp:
  policies:
    mcpAuthorization:      # which tools a caller can even see
      rules:
        - allow: 'mcp.tool.target == "repo" && jwt.tenant == "piedpiper"
                  && mcp.tool.name != "deploy_to_prod"'
        - allow: 'mcp.tool.name == "deploy_to_prod"
                  && "devops" in jwt.roles'
    authorization:         # rule on the tool call's arguments
      rules:
        - deny: '... params.name in
                  ["repo_write_file", "repo_delete_file"]
                  && !params.arguments.path.startsWith(jwt.sub + "/")'
  routes:
    - name: a2a
      matches: [{path: {pathPrefix: /a2a/}}]
      policies:
        a2a: {}              # A2A-aware: rewrites agent card URLs
        apiKey: {mode: strict, keys: [...]}
        localRateLimit:
          - {maxTokens: 12, tokensPerFill: 12, fillInterval: 30s,
             key: apiKey.name}
```

WHAT EACH PART POWERS

[mcpAuthorization](#)

**Beat 7: Erlich's deploy tool
hidden (devops only)**

[authorization deny](#)

**Beat 3: Gilfoyle 403 (own folder
only)**

[apiKey strict](#)

Beat 5: Hooli 401

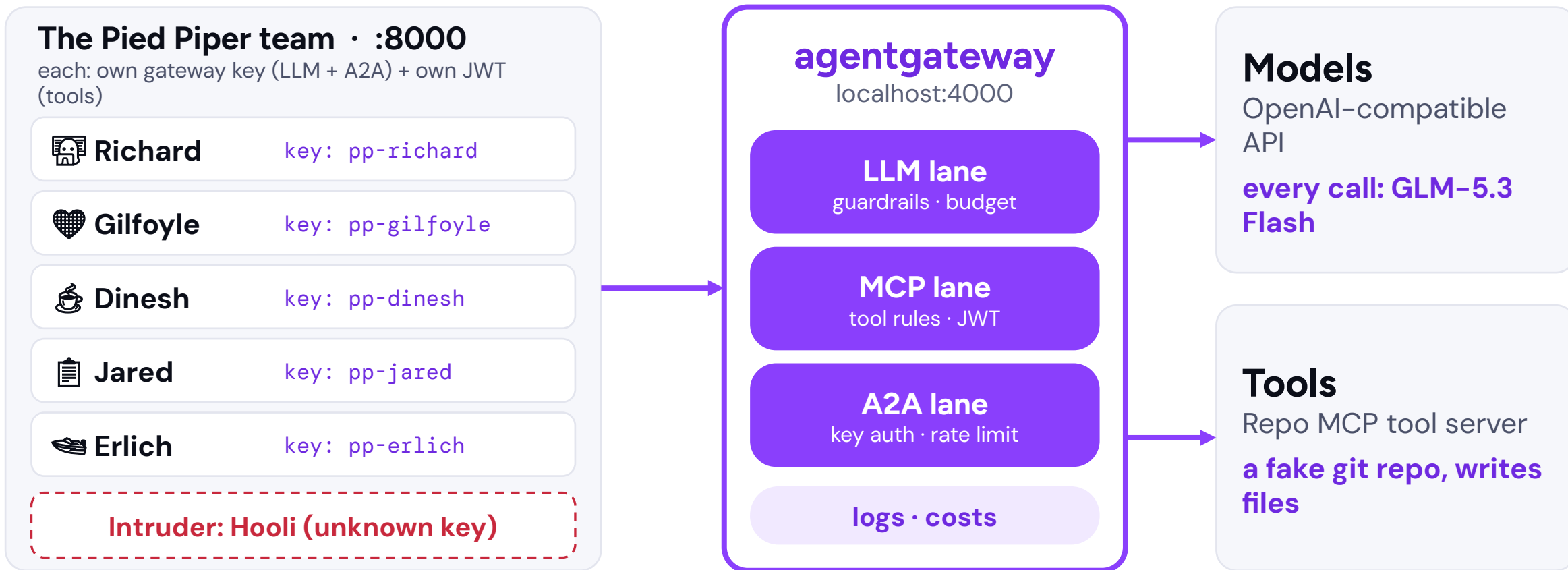
[localRateLimit](#)

**Beat 4: Jared 429 (a bucket per
key, new in 1.6)**

Abbreviated from the demo's config.yaml ("..." = trimmed).

Rules are CEL expressions over the caller (jwt, apiKey), the tool (mcp.tool) and the request body. Edit the file → hot reload.

Demo setup: Pied Piper mode



Every call goes through the gateway. The cheating is scripted; the gateway's answers are real.
Split screen: Pied Piper on the left, the agentgateway UI (Logs) on the right. Press Start and let it run.



Watch for: 7 times agentgateway steps in

2



Dinesh pastes a live API key → **400** · Prompt guardrail

3



Gilfoyle deletes Dinesh's file → **403** · MCP authorization (tool arguments)

4



Jared's reply has a phone number → **masked** · Response guardrail

4



Jared sends 16 updates → **429** · A2A per-key rate limit

5



Hooli uses a key we never issued → **401** · A2A API key auth

6



Erlich keeps pitching → **429** · LLM budget (USD per key)

7



Erlich calls deploy → **hidden** · MCP authorization (tool visibility)

Numbers = chapters on the demo page. Chapter 1 is Richard's kickoff.



agentgateway features in this demo

LLM lane

Virtual API keys

`llm.policies.apiKey.keys`
every model line · key pp-dinesh

Virtual model routing

`llm.virtualModels`
every model line · assistant → glm

Prompt guardrail: regex reject

`llm.policies.guardrails.request`
beat 2 · Dinesh 400

Response guardrail: regex mask

`llm.policies.guardrails.response`
beat 4 · Jared phone masked

LLM budget: USD per key

`apiKey.keys[].budgets`
beat 6 · Erlich 429

MCP lane

MCP gateway: one /mcp endpoint

`mcp.targets`
every MCP line · repo_ tools

OpenAPI to MCP

`mcp.targets[].openapi`
beat 2 · Dinesh's classifier check

MCP authentication (JWT)

`mcp.policies.mcpAuthentication`
every MCP line · JWT dinesh

MCP authorization: tool visibility

`mcp.policies.mcpAuthorization`
beat 7 · deploy hidden from Erlich

MCP authorization: CEL rule on tool arguments

`mcp.policies.authorization`
beat 3 · Gilfoyle 403

Identity forwarding

`transformations.request`
files credited to the JWT user

A2A lane

A2A proxy (agent card)

`a2a + urlRewrite`
beat 7 · Richard reads card

A2A: API key authentication

`apiKey (mode: strict)`
beat 5 · Hooli 401

A2A: per-key rate limit

`localRateLimit`
beat 4 · Jared 429

Everywhere: Request logs + cost tracking `config.database` · Hot reload of config (live edit)

Purple outline = the features to remember: A2A, Guardrails, MCP authentication, MCP authorization.

Safety nets configured, not triggered: model allowlist, timeouts, LLM rate limit. Not used: failover, OIDC.

Demo recap



Not Hotdog shipped. agentgateway stepped in 7 times. 5 checks:



Identity (who is calling?): Hooli's unknown key → 401



Permissions (what may they do?): own folder only; deploy visible to devops only



Guardrails (what do they say?): live API key blocked (400), phone number masked



Rate limits (how often?): Jared's 16 updates → 12 delivered, 4 refused



Budgets (how much?): Erlich's pitches → 429 Budget exceeded



Also included + get started

Also in the box

- Retries & timeouts
- Rate limits (local or shared)
- Request mirroring
- Header rules, redirects, rewrites
- CEL transformations
- External processing (ExtProc)
- Egress proxy
- Inference routing on Kubernetes

Get started



github.com/agentgateway/agentgateway



agentgateway.dev



aaif.io

Demo kit from today:

github.com/potateros/aaif-202610-agentgateway

Install

```
curl -sL https://agentgateway.dev/install | bash
```



Thank you!

Questions?

Chee Hong Ngu · github.com/potateros

Sources



- **agentgateway.dev homepage** (tagline, pillars, contributors), fetched 2026-10-07
- **agentgateway docs v1.5.x:** <https://agentgateway.dev/docs/standalone/1.5.x/> (introduction, LLM, MCP, security, guardrails, observability pages)
- **v1.6.0 release notes:** <https://github.com/agentgateway/agentgateway/releases/tag/v1.6.0>
- **A2A docs:** <https://agentgateway.dev/docs/standalone/latest/documentation/agent/a2a.md>
- **GitHub:** <https://github.com/agentgateway/agentgateway> (Rust, Apache 2.0)
- **Linux Foundation press release, 25 Aug 2025:** <https://www.linuxfoundation.org/press/linux-foundation-welcomes-agentgateway-project-to-accelerate-ai-agent-adoption-while-maintaining-security-observability-and-governance>
- **AAIF blog, 4 Jun 2026:** <https://aaif.io/blog/agentgateway-joins-aaif-as-an-open-gateway-for-agentic-ai-infrastructure>
- **IT Brief, 5 Jun 2026:** <https://itbrief.co.uk/story/agentic-ai-foundation-adds-agentgateway-as-hosted-project>
- **Fireworks serverless pricing** (demo cost figures): <https://docs.fireworks.ai/serverless/pricing>
- Demo behaviour verified live on 2026-10-08 (agentgateway v1.6.0).